

SHUBHAM GORE

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EDUCATION

University of Southern California, Los Angeles CA Jan 2024 – Dec 2025
Master of Science in Computer Science – Intro to AI, Machine Learning, Analysis of Algorithms, NLP

SRM Institute of Science and Technology, Chennai TN Aug 2019 – May 2023
B.Tech in Computer Science and Engineering - Data Structures

TECHNICAL SKILLS

Languages: Java, Python, Go, JavaScript, TypeScript, C++, C#, MySQL, SQL/NoSQL, Bash, HTML/CSS
Frameworks/Libraries: React.js, Node.js, Spring Boot, Tailwind, FastAPI, Numpy, Pandas, GraphQL, Oracle
Platform Tools: Git (version control), Docker, Kubernetes, AWS, GCP, Azure, Jenkins, Datadog, MongoDB, PostgreSQL, Atlassian, Confluence, JIRA
ML/Data Science: PyTorch, TensorFlow, Keras, LangChain, Hugging Face

EXPERIENCE

Software Development Engineer (Directed Research) Los Angeles, CA
University of Southern California Jun 2025 – Present

- Architected the backend for a complex multi-agent simulation, delivering RESTful APIs for real-time state management, event processing, and outcome determination.
- Collaborated cross-functionally to create real-time service to sync match outcomes with the Unity frontend, ensuring seamless league updates.

Machine Learning Intern Pune, India
Centre for Development of Advanced Computing (CDAC) Aug 2023 – Oct 2023

- Designed an end-to-end system for network traffic prediction, building and deploying time-series models that reduced prediction error by 8% on the GEANT dataset.
- Engineered a complete MLOps pipeline using Docker and CI/CD to automate the training and deployment of predictive models, demonstrating skills in building robust, end-to-end software systems.

Software Development Engineer Intern Pune, India
Avaya Jan 2023 – Jun 2023

- Developed scalable micro-services, REST API with Spring Boot/Nodejs and Kafka, reliably processing over 1M+ daily messages on a distributed CCaaS platform.
- Improved fault tolerance by 30% by implementing graceful shutdowns and integrating Datadog monitoring for comprehensive, real-time alerting.
- Enhanced system stability by increasing unit test coverage to over 80%, preventing production-level bugs.
- Automated deployments with Jenkins CI/CD pipeline in an Agile environment.

PROJECTS

WiM for Long Context Retrieval | *PyTorch, CUDA, TensorRT, GGUF, ONNX* Mar 2025 – Apr 2025

- Optimized LLM by developing a custom CUDA kernel, designing an efficient solution that accelerated inference.
- Achieved an improvement of 0.08 BERT score by training a PPO-based RL model for RAG long-context generation, minimized hallucinations by optimizing margin selection.
- Fine-tuned a 500M-parameter LLM with LoRA and 8-bit TensorRT quantization on HotpotQA via Hugging Face.

Full-Stack Graph RAG Chatbot | *LangChain, Backend, Neo4j, Next.js, CI/CD, OAuth, Data ingestion* Sep 2024 – Nov 2024

- Designed and deployed a full-stack application, featuring a React.js frontend and FastAPI backend that handled 10K+ documents and reduced query latency by 35% through Redis caching.
- Improved search relevance by leveraging a Neo4j graph database to capture complex relationships within 10K+ eCommerce reviews.

LEADERSHIP AND ACHIEVEMENTS

- Initiated and led a team of 4 to design and build a full-stack OCR application, taking the project from initial concept to a functional prototype achieving 93% accuracy.
- Spearheaded research and co-authored a paper on customer prediction using Neural Networks and SMOTE-ENN, published at AISP'23 (Int'l AI & Signal Processing Conference).